THE BEST THAT MONEY CAN'T BUY

BEYOND POLITICS, POVERTY, & WAR



JACQUE FRESCO

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BEYOND POLITICS, POVERTY, & WAR

BY JACQUE FRESCO

Models, Illustrations & Photos
JACQUE FRESCO & ROXANNE MEADOWS

Model Designs
JACQUE FRESCO

GLOBAL CYBER-VISIONS
VENUS • FLORIDA

Publisher's Cataloging-in-Publication
(Provided by Quality Books, Inc.)
Fresco, Jacque, 1916-

The best that money can't buy: beyond politics, poverty, & war / Jacque Fresco; models; illustrations & photos, Jacque Fresco & Roxanne Meadows; model designs, Jacque Fresco. — 1st ed.

p. cm.
ISBN: 0-9648806-7-9

1. Social prediction. 2. Social change.
3. Technology — Social aspects.
4. Technological forecasting. I. Title

HM901.F74 2002 303.49
QBI01-201428

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Copyright © 2002 by Global Cyber-Visions

Published by Global Cyber-Visions
The Venus Project
21 Valley Lane
Venus, FL 33960
U.S.A.
Phone: 863-465-0321
Fax: 863-465-1928
http://www.thevenusproject.com
tvp@thevenusproject.com

This book was set in Adobe Centaur and Adobe Serpentine
Designed by Roxanne Meadows
Composition by Jonathan Pennell

International Standard Book Number: ISBN: 0-9648806-7-9
Library of Congress Catalog Number: 2001 135379
First Global Cyber-Visions Edition
10 9 8 7 6 5 4 3 2 1
Printed in the United States of America

31143006830005
303.49 Fre
Fresco, Jacque, 1916-
The best that money can't buy : beyond politics, poverty, & war

ACKNOWLEDGMENT

MY APPRECIATION GOES OUT TO MANY other contributors who have made this manuscript possible.

Steve Doll whose untiring efforts in helping to put the manuscript in order is deeply appreciated. He also offered many relevant editorial suggestions. Steve Doll with his years of experience in writing for many technical and social publications had an immediate grasp of the concepts described in this book.

Roxanne Meadows my dearest friend and companion helped with the organizing and layout of the book while making many helpful editorial suggestions. She has devoted her time and efforts for the past 25 years towards producing renderings, models, photographs, video editing, and constructing the buildings on the 25-acre site in Venus, Florida, while fulfilling highly challenging, full-time professional assignments of her own. Without her untiring patience, effort and dedication we would not have accomplished all that has been put forth towards the foundation of The Venus Project.

Susan Bottom and Marc Ponomareff, who have both assisted greatly in editing and revising the manuscript. I am also indebted to Marc for the final proofreading.

Angeles Philolias, whose untiring support and effort in presenting the concepts of The Venus Project to others has always been a source of inspiration and encouragement to both Roxanne and myself. Since his death in 2001 we have deeply missed him.

Arthur Shostak and Marco Scorcelletti for their support and help in attempting to get this manuscript published.

Sam Laurie for his many excellent editorial suggestions and support of the project.

Millard Deutsch has been very helpful in patiently typing as I dictated portions of this book.

I thank Ingrid and Dr. David Pon, and Wilson Hawthorn for their kindness and support of the project.

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INTRODUCTION

IN OUR LONG AND EXPONENTIALLY GROWING LIST of technological achievements, perhaps none stand out so prominently as our forays beyond our own planet. With the dawn of the space age, astronauts and the public at large enjoy the unprecedented opportunity and privilege of viewing the earth in all of its colorful splendor and beauty, suspended against a backdrop of the black void of space. The narrative that usually accompanies this view is a whimsical one, reflecting the bond between earth and the billions of people who share this planet.

Global consciousness seems to inspire most of the space travelers who make these uplifting observations. For theirs has been an emotional, even spiritual experience. They seem to believe that this view from outer space cannot be interpreted as anything less than a grand global awakening to the realization that all the people who share our planet make up one, single community of brothers and sisters. They believe this viewpoint will help unite the nations of the world to build a peaceful future for humankind, both for the present generation and all those to follow.

Such has also been the preoccupation of many poets, philosophers, and writers through the ages who have bemoaned the artificiality of the borderlines that separate humankind. Despite the vision and hope of astronauts, poets, writers, and visionaries the reality is that nations are continuously at war with one another, while poverty and hunger still prevail in many places throughout the world, including the United States.

So far, not one astronaut who has arrived back on Earth with this cleansed social consciousness has offered a proposal to transcend the world's limitations and build a world where no national boundaries are represented. Each remains loyal to his/her particular nation-state, and will not venture beyond the boundaries of patriotism — "my country, right or wrong" — for to do so may risk their positions of prestige.

The problems we face in the world today are mostly of our own making. We must accept that the future depends upon us, the divine intervention of mythical characters in white robes who descend from the clouds or benign visitors from other worlds are illusions that cannot solve the problems of our modern world. The future of the world is our responsibility and depends upon the decisions that we make today. We are our own salvation or damnation. The shape and solutions of the future rely totally on the collective effort of all people working together.

Science and technology race ahead of us into the future, revealing new horizons in all areas. New discoveries and inventions appear at a rate never seen before in history. This rate of change will continue to increase in the years to come.

Unfortunately, the books and articles that attempt to describe the future have one foot rooted firmly in the past, while interpreting the future through today's concepts and technology. Most people feel comfortable with and are less threatened by this type of change. But they often recoil when confronted with proposals about changing the way they live. For this reason, when speaking of the future, very few explore or discuss changes to our social structure, much less our values. And yet that structure and those values reflect earlier times with different stresses and different levels of understanding. Any author who wants to publish wisely steers clear of such emotional and controversial issues. But we feel it is high time to step out of that box. In this book we will freely explore a new scenario of the future, one that is quite attainable and not the typical alternative of gloom and doom so often presented today.

Few can envision a social structure that enables everyone to attain a life style considered Utopian, even by today's standards — or, to take it even farther, that this lifestyle could be made available without working by the sweat of one's brow.

Yet consider the fact that, thanks to our labor-saving machines and other technological advances, the lifestyle of the middle class person today far exceeds anything that even kings of the past could have experienced.

Since the onset of the machine age, humankind has maintained a sort of love/hate relationship with its mechanical progeny. We may like what the machines do for us, but we don't like what we perceive they are doing *to* us. They take away our means of making a living, and consequently our sense of purpose, instilled through thousands of years of human enterprise in which hand labor has been the primary means of meeting human needs.

Many of us fear that as machines become more and more complex and sophisticated, and our dependence on them grows, we give over much of our own independence and become as they — passionless, unfeeling automatons whose sole purpose in life is to work, work, work. Such people fear that, eventually, these mechanical children could develop minds and wills of their own and enslave humanity to do their soulless bidding.

Many fear that our values and behavior will change to meet the dictates of uniformity, and that humanity will lose those very qualities which make us distinctly human. The purpose of this book is to explore other visions and possibilities for the future that nurture human growth and achievements as the primary goal of a healthy society. We propose to discuss the many options and roles the individual will play in the unfolding of this cybernated age, an age in which our world is rebuilt by prodigious machines and governed by computers.

Most writers of the twentieth century who attempted to present a vision of the future, blinded either by national ego or self-centeredness, failed to grasp the significance and meaning of the methods of science as they might be applied to the social system.

Although it may appear that the thrust of this book is the technology of the future, our major concern is the effect a totally cybernated world would have on humanity and on the individual. Of course, no one can really predict the future with precision. There are simply too many variables beyond our comprehension. New inventions, natural and man-made disasters, or new, uncontrollable diseases may radically alter the course of civilization. While we cannot confidently predict the future, we will most surely live it. Every action and decision we make — or don't — ripples into the future. For the first time, we have the capability, the technology, and the knowledge to direct those ripples.

When applied in a humane direction, the primary directive of the emerging, cybernated age could be the synthesizing of technology and cybernetics into a workable and human synergy for the people of all nations. It could achieve a world free of hunger, war and poverty — a world humanity failed to achieve throughout history. But if civilization continues on its present course we will simply repeat the same cycle of inadequacy all over again.

Applying what we already know to enhance the quality of all human life on earth could at the same time — indeed, must — protect the environment and the symbiotic process of living systems. It is now mandatory that we intelligently rearrange human affairs to comply with available resources. The proposals of this book are about the limitless, untapped potentials and the future application of new technologies where our health, intellect and well-being are involved, not only in the material sense but also with a deep concern for one another. Only in this way can science and technology support a meaningful and humane civilization.

Many of us who think seriously about the future of human civilization are familiar with stark scenarios of the new millennium in a world of growing chaos and disorder, of soaring populations and dwindling natural resources. Emaciated children cry out to us from decayed cities and villages, their mouths agape over bellies swollen from malnutrition and wasting disease. In more affluent areas, the problems associated with urban sprawl, air and water pollution and escalating crime take their toll on the quality of life, even for those who consider themselves sufficiently removed from these conditions. But even the financially wealthy exist at a tremendous disadvantage with their inability to grasp the damage of technology applied without social concern.

With the advances in science and technology over the last two hundred years, one may well ask: does it have to be this way? There is no question that the human application of science and technology can carry us with confidence and assurance into the future. What is needed is a change in our sense of direction and purpose. Our main problem is a lack of understanding of what it means to be human and the full realization that we human beings are not separate from nature. Our values, beliefs, and behavior are as much a part of the natural law as any other natural process. We are all an integral part of the chain of life.

In this book, we present an alternative vision for a sustainable new world civilization unlike any social system that has gone before. Although this vision is highly compressed, it

is based upon years of study and experimental research. We call for a straightforward redesign of our culture, in which the age-old inadequacies of war, poverty, hunger, debt, and unnecessary human suffering are viewed not only as avoidable, but also as totally unacceptable. Anything less simply results in a continuation of the same catalog of problems inherent in the present system.

CHAPTER 1

A DESIGN FOR THE FUTURE

THE FUTURE IS FLUID. Each act, each decision, each development adds new possibilities and eliminates others. But the future is ours to direct. In older times, change came so slowly that generations passed with minimal impact on the daily business of surviving. Social structures and cultural norms remained static for centuries.

In the last fifty to a hundred years technology and social change accelerated to such an extent, governments and corporations now consider change management a core process.

Hundreds of books address technological change, business process management, human productivity, and environmental issues. The best of our universities offer advanced degrees in public and environmental affairs. Almost all overlook the major element in all these systems — human beings and their social structures and culture. Technology, policy and automation do nothing until humans accept them and apply them to their daily lives. This book offers a blueprint to consciously fuse all these elements to create a sustainable future for all our planetary inhabitants, as well as fundamental changes in the way we regard ourselves, one another and our world. This can be accomplished through the infusion of technology and cybernetics applied with human and environmental concern to secure, protect, and encourage a more humane world for all people.

How may such a prodigious task be accomplished? First, we need a current survey of all our available planetary resources. All the discussion in the world on what is scarce and what is plentiful is just so much talk until we actually measure our resources. We must first baseline what we have around the world. This information must be compiled to ascertain the possible parameters for *humanizing* social and technological development. This may be accomplished through the use of large-scale, computer-based processors that will assist us in defining the most humane and appropriate way to manage environmental and human affairs. This is essentially the function of government. With computers able to process trillions of bits of information per second, existing technologies can far exceed the human capacity for processing the necessary information and arriving at equitable and sustainable decisions involving the development and distribution of physical resources. With this potential we may eventually surpass the need for political decisions made on the basis of power and advantage.

Eventually, with artificial intelligence, money may become irrelevant, particularly in a high-energy civilization in which a material abundance eliminates our belief in the concept

of scarcity valuation. We have arrived at a time when new innovations in the methods of science and technology can provide abundance to all the world's people. It is no longer necessary to consciously withhold efficiency by planned obsolescence, or to utilize an old and outworn monetary system

Although many of us like to consider ourselves forward-thinkers, we still cling tenaciously to all the values of the old monetary system. We accept unthinkingly a system that breeds inefficiencies and actually encourage the creation of shortages.

For example, while many concerns over environmental destruction and the misuse of technology are justified, many environmentalists draw bleak scenarios of the future based on present day methodologies and shortages. They view environmental destruction from the only point of reference they have to work with — existing technologies used wastefully and irresponsibly, primarily to support outmoded concepts and the economic imperative of sales turnover and customer appeal. Although we concede that technological development has been misdirected, the vast benefits far outweigh the negative factors involved. Only the most diehard environmental activist would turn his back on the many advances made toward the elevation of the human condition in areas such as medicine, communications, power generation, and food production.

If human civilization is to endure, it must eventually outgrow the need for conspicuous waste of time, effort, and natural resources. One area in which we may witness this is in architecture. Resource conservation must be incorporated into our structures.

With a conscious and intelligent application of today's science and technology to human and environmental concern, we can recreate the wetlands and enhance the symbiotic process between the elements of nature. This was not attainable in earlier times.

While many urban centers grapple with the daunting task of retrofitting new, more efficient technologies into their existing infrastructures, these efforts continue to fall far short of the potentials of technology. In addition to rebuilding our thought patterns, much of our physical infrastructure — industrial plants, buildings, waterways, power systems, production and distribution processes, and transportation systems — must be reconstructed from the ground up. Only then may we use our technology to overcome resource deficiencies and provide universal abundance.

If we are genuinely concerned about the environment and our fellow human beings, and if we really want to put an end to territorial disputes, war, crime, poverty, hunger, and many of the other problems that confront all nations today, the intelligent use of science and technology offers us the tools with which we can strive toward a new direction — one that could serve the well-being of all people, and not just a select few.

The purpose of all this technology is to ultimately free people from repetitive and boring jobs and allow humans to truly experience the fullness of human relationships, which has been denied to so many for so long. This will call for a basic realignment of the way we think about what makes us human. Our times demand the declaration of the world's resources as the common heritage of all people.

Perhaps at the passing of the next hundred years, future historians may look back on our present civilization as a transition point from the dark ages of ignorance, superstition, and social insufficiency — much as we view the world of a few hundred years ago. If we arrive at a saner world in which the maximum human potential is cultivated in every person, our descendants will not understand how our world of billions of human beings produced only one Louis Pasteur, one Edison, one Tesla, one Salk, and why great achievements in our age were the products of a relative few.

In looking ahead to the next millennium, and back to the dimmest memories of human civilization, we see the thoughts, dreams, and visions of humanity are limited by a system of scarcity. We are all products of a culture of deficiency, a culture that expects each confrontation and most activities to end with a winner and a loser, and funding restricts even technological development that has the potential to liberate humanity from its past insufficiencies.

We no longer have the luxury of such primitive thinking. There are other ways of looking at our lives and the world that we live in. Either we learn to live together in full cooperation or we will create our own extinction. To fully understand and appreciate this coming age, we must first understand the relationship between creation and creator: the machine and, as of this writing, that most marvelous of mechanisms — the human being.

CHAPTER 2

CHANGING VALUES IN AN EMERGING CULTURE

ANY ATTEMPT TO DEPICT THE FUTURE DIRECTION of civilization must include a straightforward description of the probable evolution of our culture without embellishment, propaganda, or national interest. We must reexamine our traditional habits of thought if we wish to avoid the consequences of not preparing for the future. It is unfortunate that most of us try to shape or envision this future within our present social framework, attempting to maintain values and traditions that reflect the past. Such superficial changes tend to perpetuate the problems of today. The challenges we face today cannot be answered by antiquated notions and values that are no longer relevant.

Imagine a new planet with the same carrying capacity as the earth, and that you are free to design a new direction for society on this planet. You can choose any shape or form. The only limitation imposed upon you is that your proposed, alternative social design must correspond to the carrying capacity of that planet. This new planet has more than adequate arable land, clean air and water, and an abundance of untapped resources. This is your planet. You can rearrange the entire social order to correspond to whatever you consider the best of all possible worlds. Not only does this include environmental modification, but also human factors, interpersonal relationships and the reordering of education.

This need not be complicated. It can be an uncluttered approach, unburdened by any past or traditional considerations, religious or otherwise. This is a prodigious project, perhaps calling for the introduction of many disciplines, applied to the way the inhabitants of your planet conduct their lives — always keeping in mind for whom and for what ends this social order is designed. Feel free to transcend present realities and reach for new and inventive ideas with which to shape your world of the future. An exciting exercise, isn't it? What we propose is nothing more nor less than applying that analysis to our planet.

To prepare for the future we must be willing to test newer concepts. This means that we must acquire enough information to evaluate these concepts, unlike the travelers who go to a foreign land and immediately compare everything with their own hometown. To understand people of another place we must set aside our expectations of behavior, lay aside the value patterns to which we have become accustomed.

If you believe today's values and virtues are absolute and ultimate, and reflect the final value systems for all times and all civilization, then you may find our projection of the future shocking and unacceptable. We must feel and think as freshly as possible about the almost limitless possibilities and combinations of life patterns humankind may explore for attaining even higher levels of intelligence and fulfillment in the future.

Although such individuals as Plato, Edward Bellamy, H.G. Wells, Karl Marx, and Howard Scott have all made some attempts to plan a new civilization, the established social order inevitably considered them impractical dreamers with Utopian designs that ran contrary to the innate elements of human nature. Arrayed against these social pioneers was a formidable status quo composed of vested interests that were comfortable with the way things were, and a populace at large that, out of years of indoctrination, went unthinkingly along for the ride. These were the millions of unappointed guardians of the status quo. The outlook and philosophy of the leaders were consistent with their positions of differential advantage.

In spite of the vast advances in the human condition achieved through the process of objective scientific investigation, and the breaking down of long-standing fears and superstitions regarding the world we live in, the world today is still not a reasonable place. Many attempts to make it so have failed due to selfish individual and national interests. Deeply rooted cultural norms that assume someone must lose for someone else to gain (scarcity at it most basic) still dictate most of our decisions. For example, we still cling to the concept of competition and inadequate compensation for people's efforts, i.e. minimum wage, when such tactics are no longer applicable or appropriate to our capabilities and resources, never mind their effect on human dignity and any possible elevation of the human condition.

At this turning point in our civilization, we find our difficulties compounded by the fact that too many of us still wait for someone, a messiah perhaps, the elusive "they", or an extraterrestrial to save us. The irony behind this is that, as we wait for someone to do it for us, we give over to them our freedom of choice and movement. We react rather than act toward events and issues.

The future is our responsibility and significant change will most probably not take place until the vast majority of the world's people lose confidence in their dictators and their elected officials' ability to solve problems. It will most likely take an economic catastrophe of unprecedented proportions resulting in an enormous amount of human suffering to bring about true social change. Unfortunately, this does not guarantee that the change will be beneficial.

In times of conflict between nations, we still default to answering a perceived threat with a threat, developing weapons of mass destruction and training people to use them against other people whom we regard as our enemies. Many of our so-called social reformers tried to solve our problems of crime within the framework of the monetary system by building more prisons and enacting new laws, i.e. gun legislation and "three times and you're out" provisions in an attempt to govern crime and violence. This has accomplished very lit-

tle, yet requests for additional funding to build more prisons and hire more policemen fare far better in legislatures and voting referendums than do pleas for additional education funding or aid to the poor. Somehow in the era of true plenty, we have meanly adopted national approval for punishment as an answer to all problems. One symptom of insanity would surely be the repeating of the same mistake, over and over again — each time expecting a different outcome. Our society is, in this sense, truly insane.

The Manhattan Project developed the first atomic device to be used against human populations — and launched the most intensive and dangerous weapons build-up in history. The Manhattan Project was also one of the largest and best-financed projects ever undertaken. If we are willing to spend that amount of money, resources and human lives in times of war, we must ask why we don't commit equal resources to improving the lives of everyone and anticipating the humane needs of the future? The same energies that went into the Manhattan Project could be channeled to improve and update our way of life, and to achieve and maintain the optimal symbiotic relationship between nature and humankind.

If our system continues without the appropriate modification and environmental and social concern we will eventually face an inevitable economic and social breakdown of our outworn monetary and political system. When this occurs the established government will most likely enact a state of emergency or marshal law to prevent total chaos. I do not advocate this but without the suffering of millions of people it may be almost impossible to shake our complacency with our current way of life.

OUT OF THE DARK AGES

Many scientists participating in the space program face different challenges. For example, space scientists must develop new ways of eating in outer space. Astronauts' clothing must withstand the vacuum of outer space, enormous temperature differentials and radiation — yet remain light in weight and permit a high degree of flexibility. This new clothing design even calls for the possible development of self-repairing systems. Their challenge is to conceive of common items in completely new ways. In space, for example, clothing no longer functions as just body covering and adornment. It becomes a mini-habitat.

The space age represents a prime example of the relentless search for newer and better ways of doing things. As scientists continue to probe toward the limits of our universe, they must generate newer techniques and technologies for unexplored frontiers and never-before-encountered environments. If they cling tenaciously to the concepts of their earlier training, our explorations will fail. Had our ancestors refused to accept new ideas, the physical sciences would have progressed little beyond the development of the covered wagon.

Many young engineers, scientists, and architects face this dilemma. Bold and creative, they exit institutions of higher learning and step out into the world eager for change. They set out with great enthusiasm but are often beaten back and slowed by the established institutions and self-appointed guardians of tradition. Occasionally, some men and women

break away from traditional concepts and set out as innovators. They meet such tremendous resistance by antiquated building codes and other restrictions that their daring concepts are soon reduced to mediocrity.

Many of the dominant values that shape our present society are still medieval. The myth that we live in an enlightened age or an age of reason really has little basis. We are overwhelmed with valid information concerning ourselves and our planet, most of which we have no inkling of how to apply. Most of our customs and modes of behavior are locked in the dungeons of the mental Dark Ages and have been handed down to us.

It was difficult for early forms of life to crawl out of the primordial slime without dragging some of it with them. So it is with any entrenched value system. The most appropriate place for traditional concepts should be in a museum or in books about the history of civilization.

The most significant changes of the twenty-first century will reveal what most people never suspected, that the majority of us have the potential of people such as Leonardo da Vinci, Alexander Graham Bell and Madam Curie, if raised in an environment that encourages genuine individuality and creativity. This includes all the other characteristics believed to be the product of a special and privileged heredity of great men and women.

Even in today's so called democratic society considerably less than 4% of the world's people have supplied us with the scientific and artistic advances that sustain the operation of the world's social system.

SHAPING HUMAN VALUES

The human beings of the future, although similar in appearance, could differ considerably in their outlook, values and mindset. The social orders of the past that have prevailed into the twenty-first century consistently sought to generate loyalty and conformity to established institutions as the only means to sustain a workable society. Countless laws — usually passed after a misdeed has occurred — have been enacted in an attempt to govern the conduct of people. Those who do not conform are ostracized or imprisoned.

During the preceding eras of social change, many social reformers and those disdainfully called agitators by their detractors were not generally angry, maladjusted individuals. They were often people with a deeper sensitivity and concern for the needs of others, who envisioned a better life for all people. Among these we may count the abolitionists, advocates for woman's suffrage and child labor laws, those who practiced non-violent resistance to oppression, and so-called "freedom fighters" everywhere.

Today we accept without question most of the achievements of these reformers, who faced violent opposition, imprisonment, ridicule and even death from vested interests and the established order. Unfortunately most people are completely unaware of the identities of those individuals who helped pave the way toward social enlightenment.

Most of our parks have statues of warriors and statesman, while any monuments to the really great social innovators are rare indeed. Possibly when the history of the human race is finally written, it will be told from the viewpoint of individuals who found themselves in an alien and primitive culture and sought change in a world with a built-in tenacity to maintain things as they are.

Conformity on the part of the population makes the control of society much easier for its leaders. Our leaders emphasize the freedom that democracy provides while supporting an economic structure that imprisons it citizens under more and more debt. They proclaim that all have the opportunity to work their way to the top of the ladder through the relentless effort of individual initiative and incentive. To appease those who work hard and do not achieve the good life, religion is there to assure them that if not in this life they will obtain it in the next.

Our habits of thought and conduct reflect the effectiveness of the constant and unrelenting propaganda of radio, television, publications, and most other media. They are so effective that the average citizen is not even insulted when being categorized as a consumer — as if a citizen's sole worth to society was as a user of goods. These patterns, however, are gradually being modified and challenged by the silent infusion of the Internet and the World Wide Web.

Most people today simply expect that our televisions, computers, communications systems, and methods of production and delivery of services — even our concepts of work and reward — will continue to improve without any disruption or distress within our present value systems. But this is not necessarily so. Our dominant values with their emphasis on competition and scarcity limit our continued progress.

The most disruptive period of a transition from an established social order to an emergent system result when people are not prepared emotionally or intellectually to adjust to change. People cannot simply erase all the belief systems and habits acquired during previous times, which they associate with self-identification. Sudden changes in values without the necessary preparation will cause many of the unprepared to lose their sense of identity and purpose, isolating them from a society they believe has passed them by. Another limiting factor that prevents people from properly evaluating alternative social proposals is the lack of understanding of basic scientific principles and the factors responsible for shaping culture and our behavior.

The conflicts today with our fellow human beings are over opposing values. If we manage to arrive at a saner future civilization, the conflicts will be against problems common to all humans. In a vibrant and emergent culture, rather than having conflicts between nations, the challenges we will face will be overcoming scarcity, restructuring damaged environments, creating innovative technologies, increasing agricultural yield, improving communications, building communications between nations, sharing technologies, and living a meaningful life.

WORK AND THE NEW LEISURE

From early civilizations to the present day most humans had to work to earn a living. Most of our attitudes about work may be a carry-over from these earlier times. In the past (and still in many low-energy cultures), it was necessary for people to fetch water and carry it to their dwelling places. They gathered wood to prepare fires for heating and cooking, and fuel to burn in their lamps. It would have been very difficult — and still is for some — to imagine a time when water would rush forth in their own dwelling with the turn of a handle; to press a button for instant light would have seemed to be within the realm of magic. People of ancient times probably wondered what they would do with their time if they did not have to engage in these burdensome tasks that were so necessary to sustain their lives. In most developed countries of the world today tasks that were once so vital to a people's very survival are no longer necessary, thanks to modern technology.

Today people attend schools to acquire the marketable skills that enable them to earn a living in the "work-a-day" world. Recently, this cherished belief that one must work to earn a living is being challenged. Working for a living to supply the necessities of life may soon be irrelevant due to the ability of modern technology to provide most of these needs. As a result, many jobs have gone the way of the iceman and the elevator operator. Perhaps we have a semantic problem with the word "work." The idea of "freedom from work" should be equated with the elimination of repetitive and boring tasks that hold back our intellectual growth. Most jobs, from blue-collar assembly workers to professionals, entail repetitious and uninteresting tasks. All human beings possess a tremendous, untapped potential that they will finally be able to explore when they are free of the burden of having to work to earn a living.

At present there are no plans in government or industry to make the necessary economic adjustments to deal with the issue of the displacement of people by automated technology. It is no longer the repetitious work of just the laborers that cybernation is able to phase out but also many other vocations and professions. Engineers, technicians, scientists, doctors, architects, artists, and actors will all have their roles altered, sometimes drastically. Therefore, it is absolutely imperative that we explore alternatives to improve our social constructs, beliefs, and quality of life to secure and sustain a future for all people.

CHAPTER 3

LANGUAGE OF RELEVANCE

OF THE MANY OLD, ENTRENCHED BARRIERS to positive change, communication has proven the most intractable. The languages of the majority of the world's peoples evolved over periods of centuries, through ages of scarcity, superstition and social insufficiency, and are continuing to evolve. However, the current state of language often lends itself more to ambiguity and uncertainties where important issues are at stake, rather than to a precise and universally intelligible means of conveying knowledge. It is often extremely difficult for the average person — and even those considered above average, including leaders of nations — to share ideas with others whose worldview may be at considerable variance with their own. Also, due to semantic differences and our different experiences, words have varying shades of meaning.

What will happen should we make contact with an alien civilization, when we haven't been able to make contact with our fellow human beings? We are not ready for that. We haven't yet learned to resolve international differences by peaceful methods. Unfortunately we live in a world today where peace is simply a pause between wars.

Even in the United States, supposedly the most technologically advanced country in the world, we lack a unified, definitively stated direction. And so our policies and goals are fragmented, contradictory. The Democrats cannot communicate meaningfully with the Republicans. Elsewhere, the Israelis oppose the Arabs, the Irish Catholics clash with the Irish Protestants, the Serbs with the Muslims. Everywhere there is interracial disharmony, the inability of husbands and wives to communicate with each other or their children, labor and management strife, the communists differing with the capitalists.

How then in the name of sanity could we hope to establish any sort of meaningful level of communication with an alien civilization, with beings possessing the intelligence and social coherence to develop technologies far in advance of our own? Perhaps from the aliens' point of view, they might well wonder whether there really is intelligent life on earth.

Most world leaders seek to achieve greater communication and understanding among the nations of the world. Unfortunately, their efforts to achieve that goal have met with little success. One reason for this is that each comes to the table determined to achieve the optimal position of advantage for their own national interest. We talk a great deal about global development and global cooperation. But the "global" in each case reflects the individual nation's interests and not those of all the people of the world.

In addition we are trapped within old ways of looking at our world. While most of us agree change is necessary, many limit the extent of change if it threatens their position of advantage, just as on a personal basis they seek change in others, rather than themselves.

Many of us lack the skills to truly communicate logically when we are emotionally invested in an outcome. If a person or group of people has difficulty in communicating a point in question, rather than seek clarification they raise their voices. If this does not produce the desired effect, they may incorporate profanity or intimidating language. If this still does not work, they may resort to physical violence, punishment or deprivation as a means of achieving what they consider to be the desired behavior. Unfortunately, in some instances deprivation of the means of earning a living has been, and continues to be, used.

These tactics have never contributed to a heightened level of understanding. In fact, many of these attempts to control behavior actually increased violence and drove the parties farther apart. It would probably be difficult for a future historian to understand why the language of science and technology was not incorporated into everyday communication.

THE LANGUAGE OF SCIENCE

Ambiguity may enable lawyers, preachers and politicians to function, but it falls short in accomplishing the tasks of building bridges, dams, power projects, flying machines, or enterprises involving space travel. For these precise functions, we need the language of science. Even amid this labyrinth of ambiguity, this much more serviceable language is coming into use throughout the world, particularly in the technologically advanced countries.

If communication is to improve, we need a language with a high degree of physical correlation with the environment and human needs. We already have such a language shared by the science and technology communities and easily understood by many of the world's nations.

In other words, it is possible to establish a coherent means of communications without ambivalence. If we apply the same methods used in the physical sciences to psychology, sociology and the humanities, a great deal of unnecessary conflict could be resolved. In such disciplines of science as engineering, mathematics, chemistry, and other technical fields we have examples of the nearest thing to a descriptive universal language that leaves little room for individual interpretation.

For instance, if a blueprint for an automobile were to be given to any technologically developed society anywhere in the world, regardless of their political or religious beliefs the finished product would be very similar to that in other areas receiving the same blueprint.

The language used today by the average person is inadequate for resolving conflicting ideas, whereas the language of science is relatively free of the ambiguities and conflict prevalent in our everyday, emotion-driven language. It was deliberately *designed* — as

opposed to evolving through centuries of cultural change — as the most appropriate way to state a problem in terms that are verifiable and readily understood by most people.

Many of the great technical strides made in the present world would have been unattainable without this type of improved communication. Without a common, descriptive language, we would have been unable to prevent diseases, increase crop yields, talk over thousands of miles, or build bridges, dams, transport systems, and the many other technological marvels of this computerized age.

Unfortunately, the same does not hold true for conversational language. Any attempt to discuss or evaluate newer concepts in social design is greatly limited by our habit of comparing newer concepts to existing systems and beliefs.

IT'S A SEMANTIC JUNGLE OUT THERE

Utopian ideals have almost certainly existed as long as humans have dealt with problems and reflected upon a world free of them — or at least having only a minimal amount. The framers of scriptural references to Eden, Plato's *Republic*, H. G. Wells' *Shape of Things to Come*, and such concepts as socialism, communism, democracy, and the ultimate expression of bliss, Heaven, have all shared this Utopian dream. All attempts at creating such a world seem to have fallen far short of their vision, possibly because the dreamers and visionaries who have projected their Utopian concepts did so mostly within the framework and values of their existing culture. The language that they used was limited and subject to a wide range of individual interpretations.

When we read and discuss new ideas, the information is automatically filtered through previous experiences and patterns of associative memory. In many instances what we end up with is something other than the designers intended. Unfortunately we live in a linguistic and semantic jungle. The use of language that we inherited is insufficient and lacks the fundamental means whereby ideas can be shared.

To cite an historical example, when presented with the possibility of the transition from conventional aircraft configuration to the flying wing concept during World War II (now employed very effectively in the B-1 Stealth bomber), most people first noticed the absence of the tail assembly. This new configuration, so far removed from the conventional, made them uncomfortable and their reactions were generally hostile. Even technical people questioned the lack of stability believed to be inherent in the flying wing.

They responded with doubt and hostility. Had they used the appropriate language of investigation, they would instead have asked the designer how he intended to overcome the limiting factors in the earlier designs. The designer would have responded by presenting new ideas through design specifications and — better yet — working models of the subject matter under discussion.

To continue a discussion of the redesign of a culture — not Utopian, but simply in accordance with the knowledge and resources we have at hand — we must learn to outgrow

our egos in exchange for constructive dialogue rather than debate. In addition we must be capable of stating problems and proposing alternative solutions clearly and succinctly, in ways that are not subject to distortion of meaning or misunderstanding, even if these solutions are radically opposed to the accepted norms.

CHANGING LANGUAGE

Language evolves along with a people and their culture. With the development of newer technologies our everyday language changes accordingly. But today our technology and culture is so pervasive, we need a much more uniform meaning throughout the world. This would be similar to mathematics in that it avoids semantic differences in interpretation. This new language could have symbols designed for a close approximation with real events in the physical world. An advanced descriptive language will most likely eventually be designed by artificial intelligence and continuously updated to remain relevant to existing circumstances and new situations.

Perhaps the future language may transcend words as we know them to become a series of sounds sequentially arranged to produce a desired response in others. *Language is an attempt to control behavior through the transfer of information, valid, invalid, or even irrelevant to the situation.*

Perhaps as it becomes essential that clarification of goals be stated precisely, our language will undergo considerable modification. The refinement and enlargement of our present language cannot be explained within the bounds of our existing temporary idiom usage. It must evolve and undergo constant refinement, while increasing the scope of meaning before it becomes an effective means of communication between people.

It is unfortunate that Guttenberg invented the printing press before the English alphabet and spelling had become stabilized. Many of our language's idiosyncrasies still endure from those early days of experimentation. No sophistication of phrasing or vocabulary alter the fact that different words, and even the same words in different sequences, have so many possible interpretations that their semantic connotations differ from sender to receiver — and from that receiver to others. Our language has an amazing richness and flexibility that easily accommodates change. But when the exception is the rule, clear communication is a challenge.

In the future, people assisted by computers could initiate a language that will offer closer areas of understanding and a much simpler structure with fewer guttural sounds. For example, a series of signals combining acoustical, optical, olfactory, and teletactile electronic patterns will tell a story in seconds, rather than in many sentences or pages.

Such a methodology is not far removed from that which enables fish to find the Oronoco River thousands of miles away from their starting point without ever having been there before. Fish have receptors that sense the earth's magnetic field, which to a large extent shapes their behavior. In a like manner, imprinting in a bird probably elicits the nest-building pattern. When our technologies are more closely aligned with natural law, airplanes

could use geomagnetic fields for navigation, just as these fields may serve as guides to migratory patterns for birds.

A clearer, more efficient means of communicating ideas and experiences would result in a more exact expression for human verbal interaction. It could open an entirely new area of science — a science of significance and meaning. The introduction of a refined language could result in a rearrangement of the associative systems in the human brain, resulting in greater understanding and reduction in conflict.

BRIDGES OVER TROUBLED WORDS

Taken in its common context, a myth is a concept or tale unsubstantiated by factual proof. For our purposes, the word denotes a non-referential method for stating problems. In this context, the author considers the matter of resolving conflicts on the basis of mutual "understanding" a myth.

For example, the possibility of members of the Jewish faith resolving the conflict of the Nazi point of view through the free exchange of views is extremely remote, if not non-existent. The same difficulties may be encountered by a well-educated African-American attempting to resolve conflict with a white supremacist organization, or a scientist sharing the Darwinian theory of evolution with religious fundamentalists. This is largely due to the fact that human beings, as yet, are not rational beings in any sense of the word.

Our values pertaining to right, and wrong, good and bad, are the byproducts and operating procedures of older social systems. For example, such slogans and catch phrases as "God is on our side", "Think American", "successful person", "well adjusted", "mature outlook", and "sharing ideas", are all statements and assessments relative to the culture in which they originated. If we genuinely hope to bridge differences, we need a much more precise language and a mindset more open to new ideas.

Actually, there is no sharing of relevant values — no communication at all — if the parties either have no common starting point or are unwilling/unable to conceive of any experience outside their own. If a person believes that it is impossible to build a flying machine, the builder of the flying machine cannot share any knowledge relative to the idea because the doubting party offers no inquiry as to how it can be accomplished. He or she has already dismissed it in their mind.

How, then, in a culture-bound society with equally limited language and ideals, can we introduce concepts of a future society with values far removed from the present? How can we orient the listener to new concepts for which, even if they have the desire to learn, they have no point of reference in their experience and thinking ability?

For the most part, we live in a perpetual "show me" state. When Nikola Tesla first introduced the concept of the wireless, there was no common understanding of the methods and dynamics of wireless transmission. Instead, Tesla oriented the uninformed through a demonstration of the working processes.

In like manner films, books, seminars, videos and ultimately a first working prototype of a new city system based on a new social direction will be essential to demonstrate the validity of our proposals.